

CareerGPS Agent

Project Name:

CareerGPS Agent

Team:

- Aalia Khan
- Aarvi Dalvi
- Ayesha Sultana

Domain:

EdTech & Future Skills

Problem Statement

Students often know which career they want, but have little visibility into how prepared they actually are for it.

For students in tier-2 and tier-3 colleges in India, access to personalised placement guidance can be limited. They may not know which skills employers value most, where their resume falls short, what to work on first, or which opportunities are realistic for their current level.

This leads to unfocused learning and applications, with students often spending time on the wrong skills or applying without understanding their readiness.

Solution

CareerGPS Agent is a career-readiness platform that turns a student's resume and target role into a practical job-preparation plan.

A student uploads their resume, selects a target role and sets the amount of time they have. The system evaluates their current skills against the requirements of that role and produces four things:

- A transparent readiness score with a breakdown of where points are gained or lost
- A week-by-week plan focused on the most important gaps
- Ranked job opportunities based on the student's current profile
- A personalised message they can send to a recruiter
- The experience is available in **English, Hindi and Marathi**, making the service more accessible to students across different backgrounds.

How It Works

The agent works through six specialised functions:

Function	Purpose
Resume Parsing	Extracts education, skills, projects and experience from the student's resume
Role Lookup	Identifies the skills required for the student's target role
Gap Analysis	Compares the student's profile with those requirements and calculates readiness
Roadmap Builder	Converts the most important gaps into a week-by-week plan
Job Matching	Ranks opportunities according to the student's existing skills
Outreach Drafting	Creates a short, personalised message for a relevant recruiter

The agent determines which function it needs at each stage, uses the result to inform the next step, and can recover when a tool returns an error or requires clarification.

What Makes It Agentic

CareerGPS Agent is not simply a chatbot that gives career advice.

The model decides which tools to use and in what order. The outputs from one step become inputs for the next, allowing the system to move from analysing a student's profile to producing concrete actions.

For example:

Resume → Role Requirements → Skill Gap → Roadmap → Job Matches → Recruiter Outreach

If a tool encounters an error, the result is returned to the agent so it can correct the request and continue rather than stopping the workflow.

The final output is therefore not just a conversation. It is a **readiness assessment, action plan and set of job-application materials**.

Readiness Score

The readiness score is deliberately kept separate from the AI's judgement.

The score is calculated using fixed Python logic so that it is **consistent, explainable and reproducible**.

- **80 points** are based on the required skills for the selected role
- **20 points** are based on relevant projects
- Skills receive full or partial credit depending on the evidence in the student's resume
- Each point can be traced back to a specific skill or project

This allows AI to handle tasks that require interpretation, such as understanding resumes and writing recommendations, while deterministic code handles scoring and other measurable outputs.

Student Outcome

Within a single workflow, a student moves from:

"Am I ready for this role?"

to:

"Here is where I stand, what I am missing, what I should work on this week, which opportunities I can target, and how I can approach them."

Bharat Impact

CareerGPS Agent is aimed at students who may not have access to dedicated career mentors or affordable one-to-one counselling.

The immediate value is individual: giving students a clearer and more affordable way to understand their career readiness.

The longer-term opportunity is institutional. Placement cells could use the same system to assess and guide large groups of students, helping colleges identify common skill gaps and provide more targeted preparation before recruitment season.

Supporting **English, Hindi and Marathi** also allows the product to serve students beyond an English-only career guidance model.

Technology

- **Backend:** Python
 - **API:** FastAPI
 - **AI:** LLM-based tool-calling agent
 - **Live LLM:** Groq
 - **Resume Processing:** PDF extraction with AI and rule-based fallback
 - **Agent Tools:** Python
 - **API Streaming:** Server-Sent Events
 - **Testing:** Pytest
 - **Deployment:** Render
 - **Containerisation:** Docker
-

Current Prototype

The live prototype accepts:

Resume PDF + Target Role + Available Weeks + Preferred Language

and returns:

Readiness Score + Skill Breakdown + Weekly Roadmap + Ranked Opportunities + Recruiter Outreach

The prototype currently supports eight entry-level roles and includes 20 sample job openings. The job dataset is intentionally kept separate from the scoring system so that it can later be connected to a production jobs API.

Live Demo:

<https://careergps-agent.onrender.com/>

API Documentation:

<https://careergps-agent.onrender.com/docs>

Git Hub Link:

<https://github.com/aarvidalvi4/careergps-agent>
